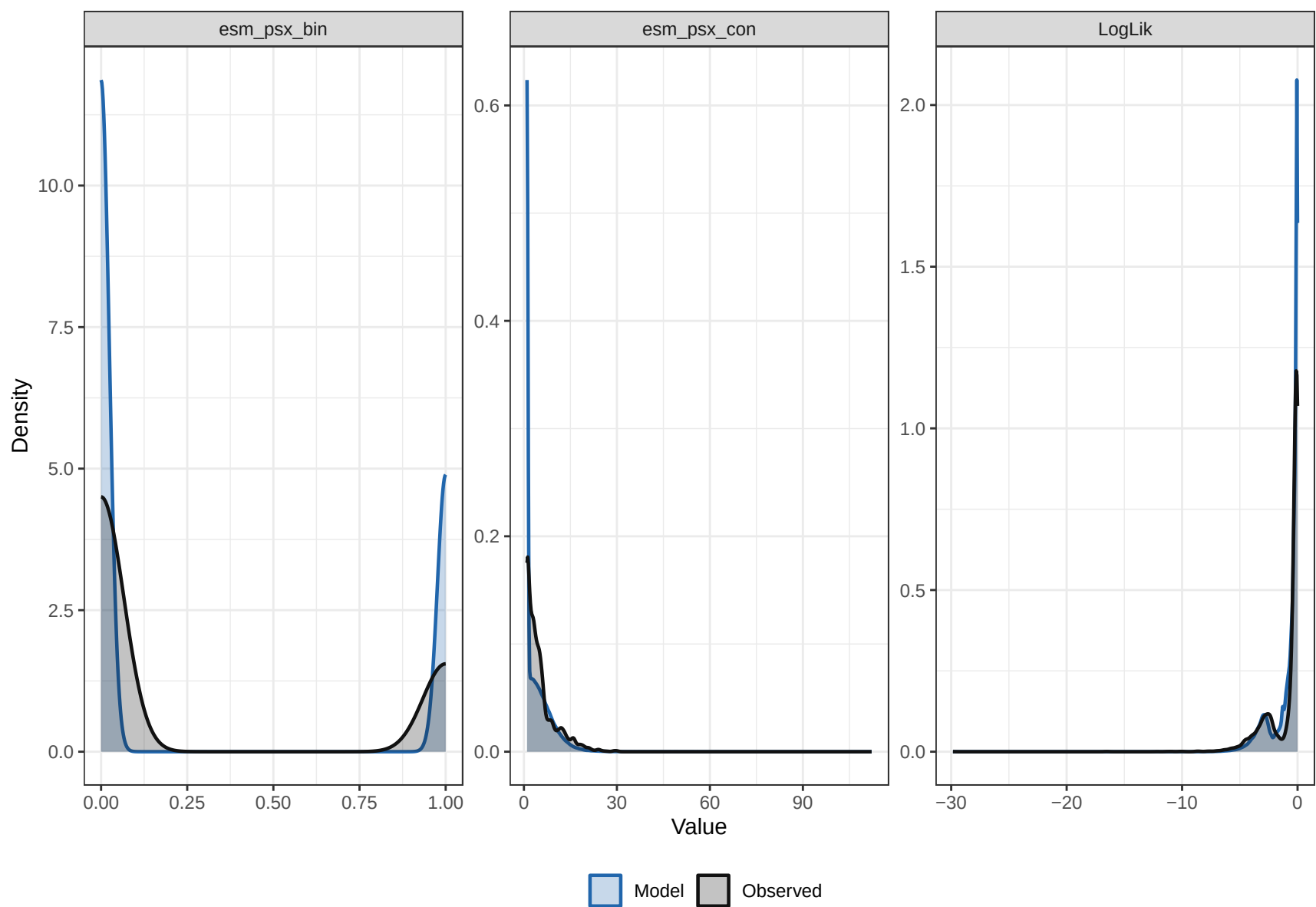
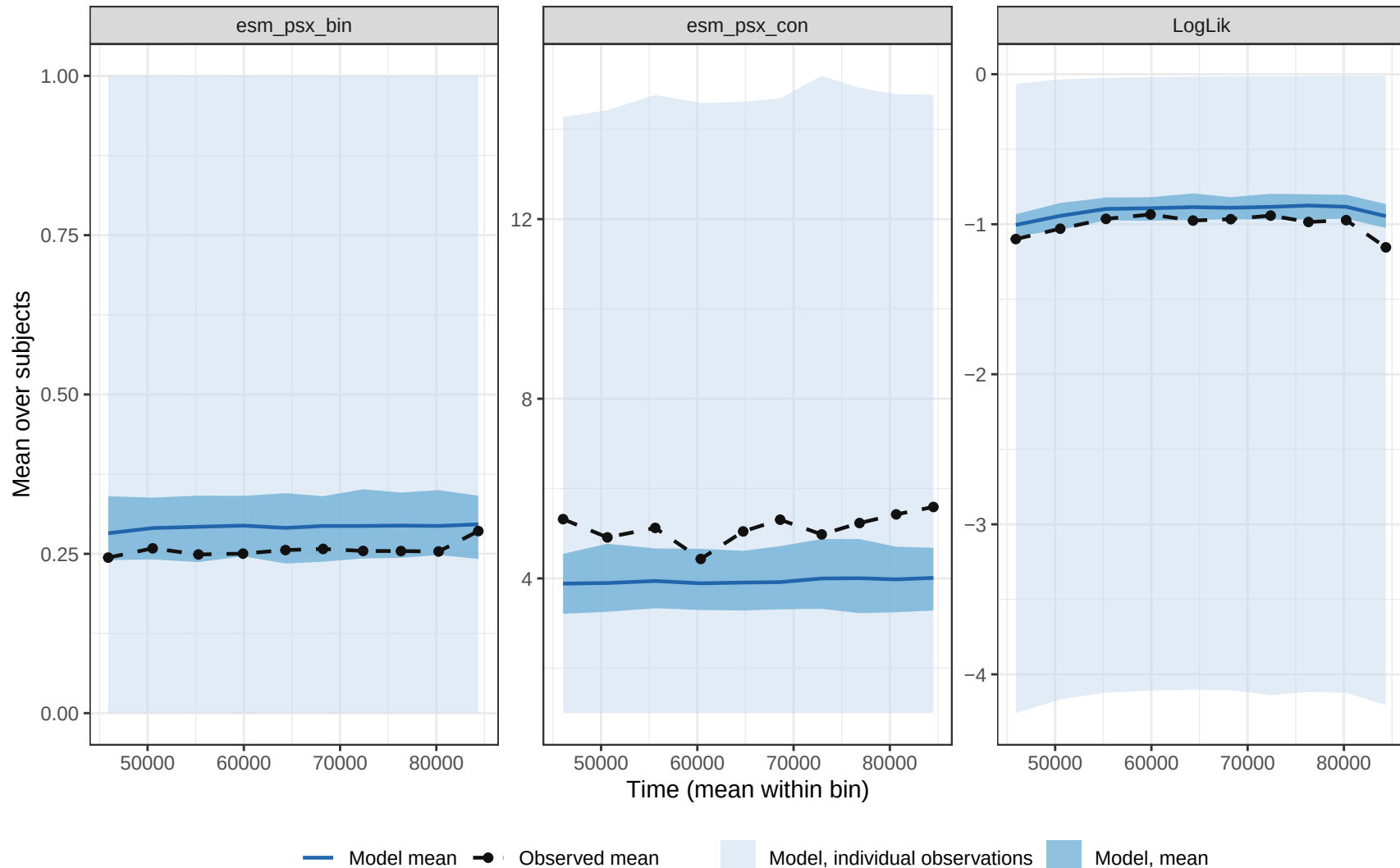




Marginal distribution of each variable. Blue: values generated from the fitted model (200 draws). Black: the observed data. A shift in location or width means the model does not reproduce the marginal, before any question of dynamics arises. LogLik is each row's log-likelihood, and its distribution should match like any other quantity -- but the observed side is evaluated at the point estimate, so it carries no parameter uncertainty while the generated side does.

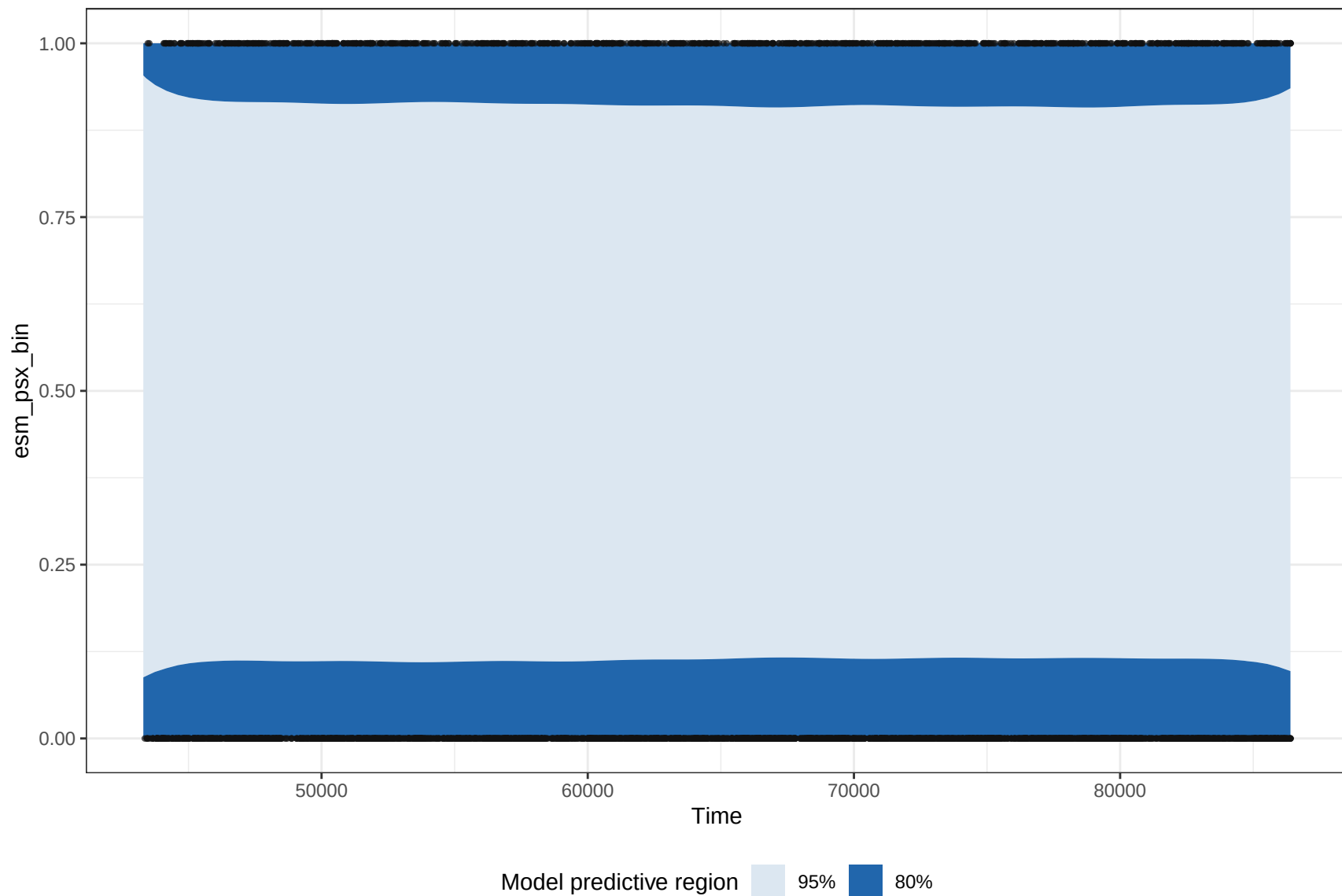
10 equal-count bins of time, 192–766 observations each



Mean over subjects at each time, in equal-count bins of time. Dark band: where the model puts 95% of the bin MEAN, across 200 generated datasets -- the reference the observed mean should sit inside. Light band: where it puts 95% of individual observations, which is the spread in the data rather than uncertainty about its mean. The mean band is much the narrower of the two, which is the point of the panel: a bias in the average level that disappears inside the observation cloud of ValueByTime is plain against it. LogLik is each row's log-likelihood, and its distribution should match like any other quantity -- but the observed side is evaluated at the point estimate, so it carries no parameter uncertainty while the generated side does.

esm_psx_bin

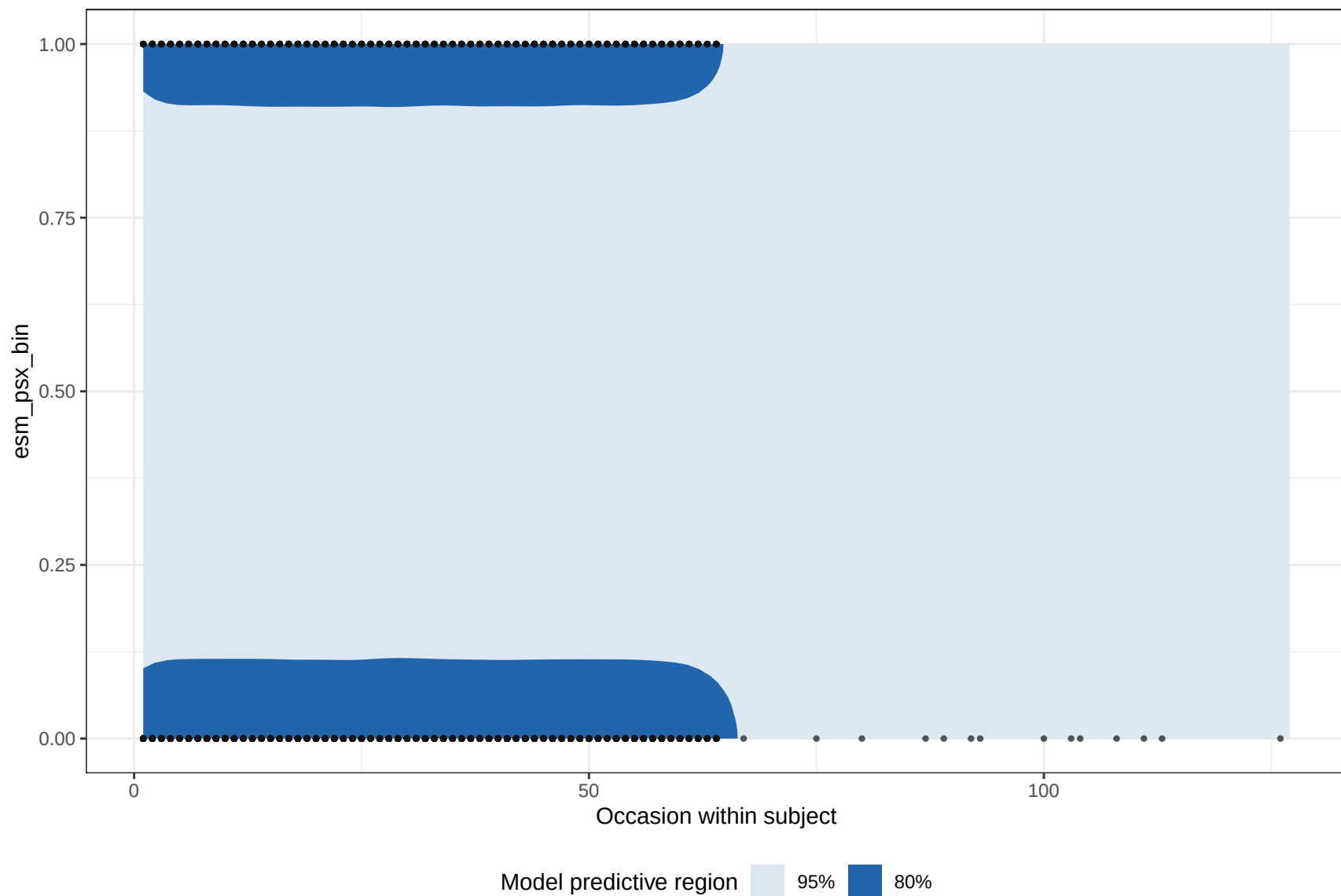
Points: observed (4000 of 7555 shown). 0% fall outside the 95% region (orange); about 5% should.



Shaded: where the model puts 50, 80 and 95% of its predictive mass, from 200 draws. Points: the observed data, orange outside the 95% region. A systematic drift of the points off the regions means the trend is wrong; regions much wider or narrower than the scatter mean the variance is.

esm_psx_bin

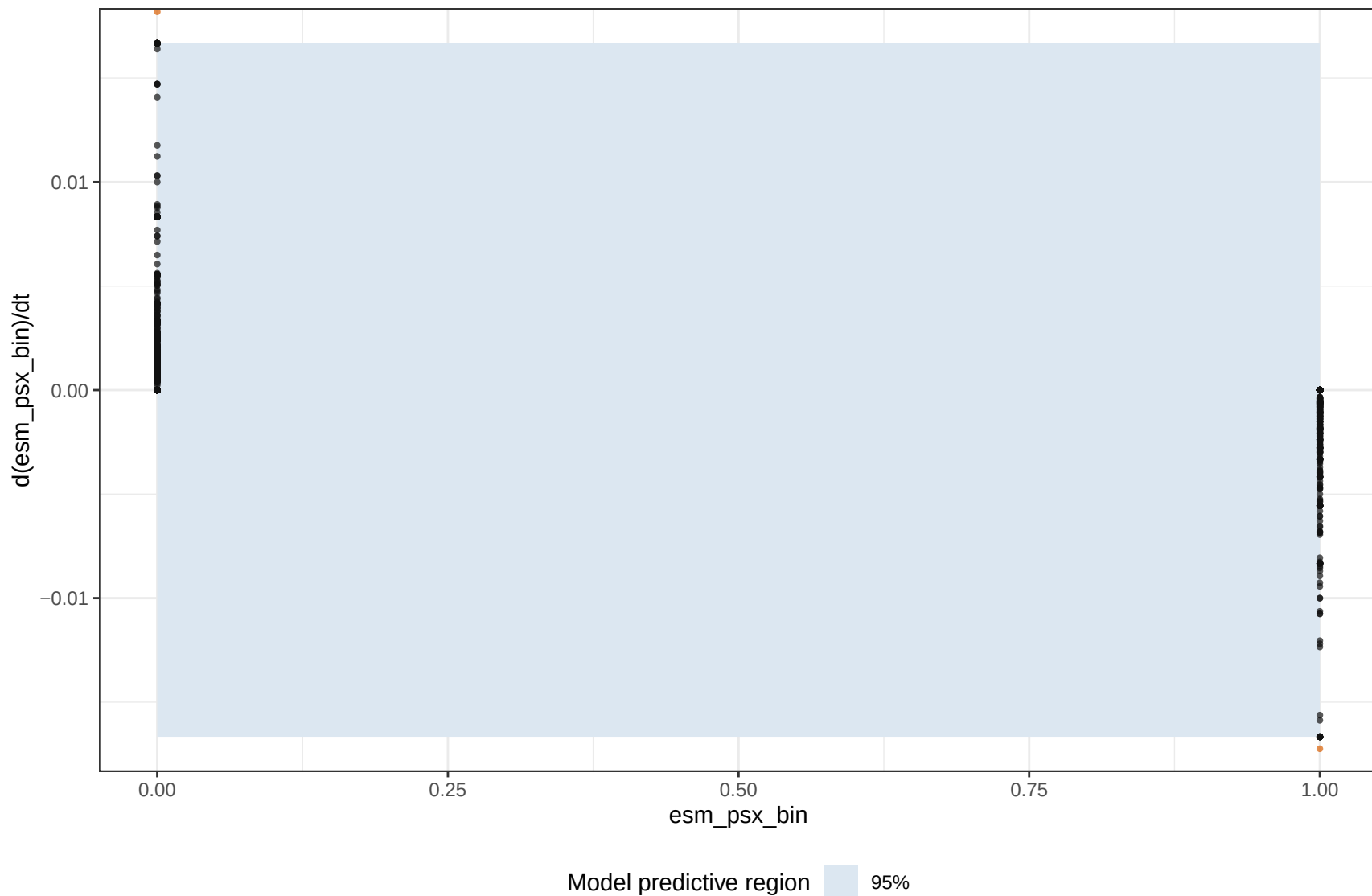
Points: observed (4000 of 7555 shown). 0% fall outside the 95% region (orange); about 5% should.



As the time panel, but against occasion number within subject rather than elapsed time. Separates a misfit that follows measurement order (fatigue, practice, panel conditioning) from one that follows time.

esm_psx_bin, lag 1

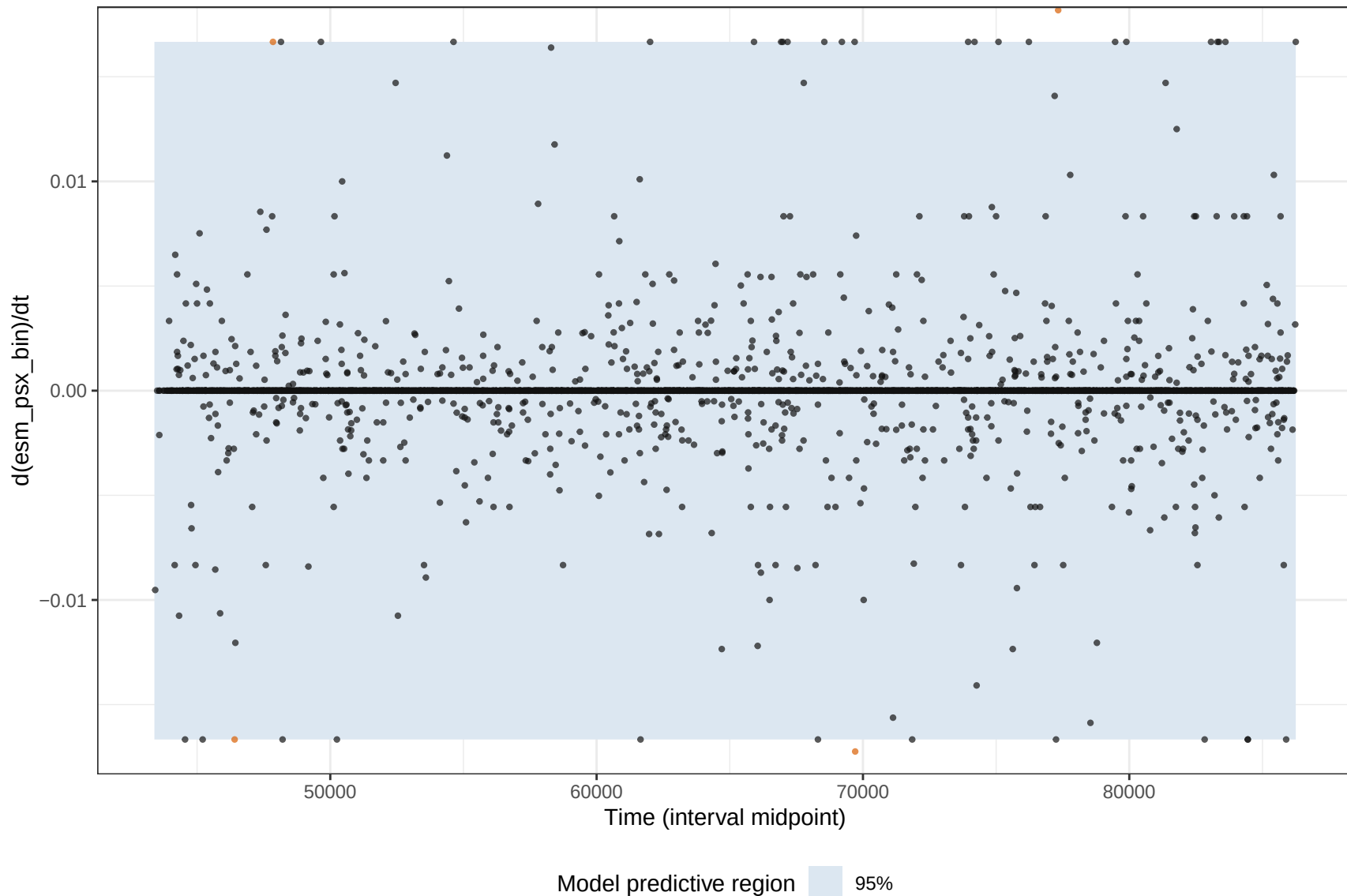
Points: observed (4000 of 4739 shown). 1.2% fall outside the 95% region (orange); about 5% should.



Phase plane: rate of change against the value it started from, shaded by the 50, 80 and 95% predictive regions. A linear model implies a band sloping down at the drift rate, the same everywhere. Observed points that curve, or that flatten at the extremes while the regions do not, indicate nonlinear dynamics. See `ctPhasePortrait()` for the model side of this on its own.

esm_psx_bin, lag 1

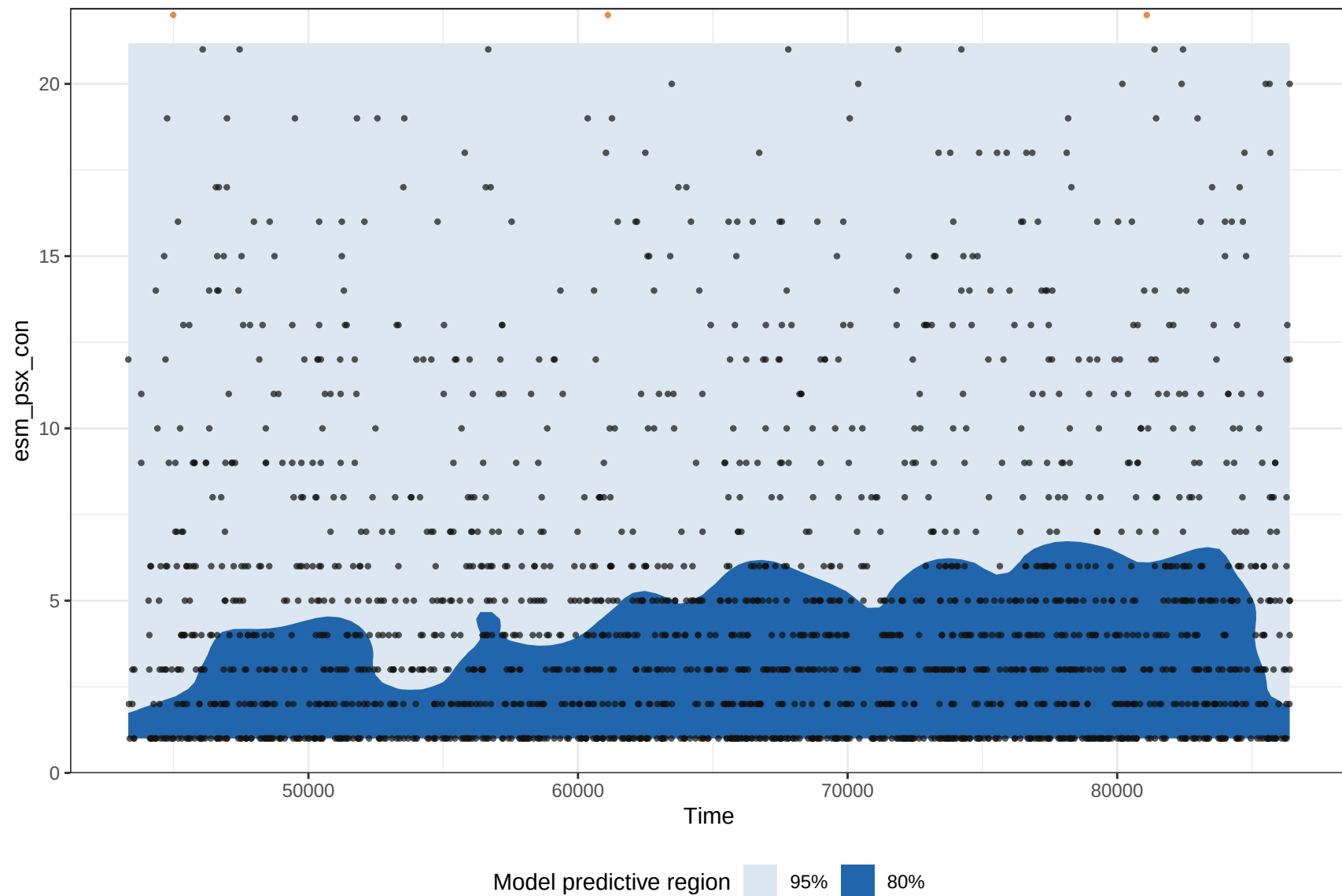
Points: observed (4000 of 4739 shown). 1.2% fall outside the 95% region (orange); about 5% should.



Rate of change against time, plotted at the midpoint of each interval and shaded by the 50, 80 and 95% predictive regions. A scatter that widens or narrows over time while the regions do not points at the diffusion rather than the drift.

esm_psx_con

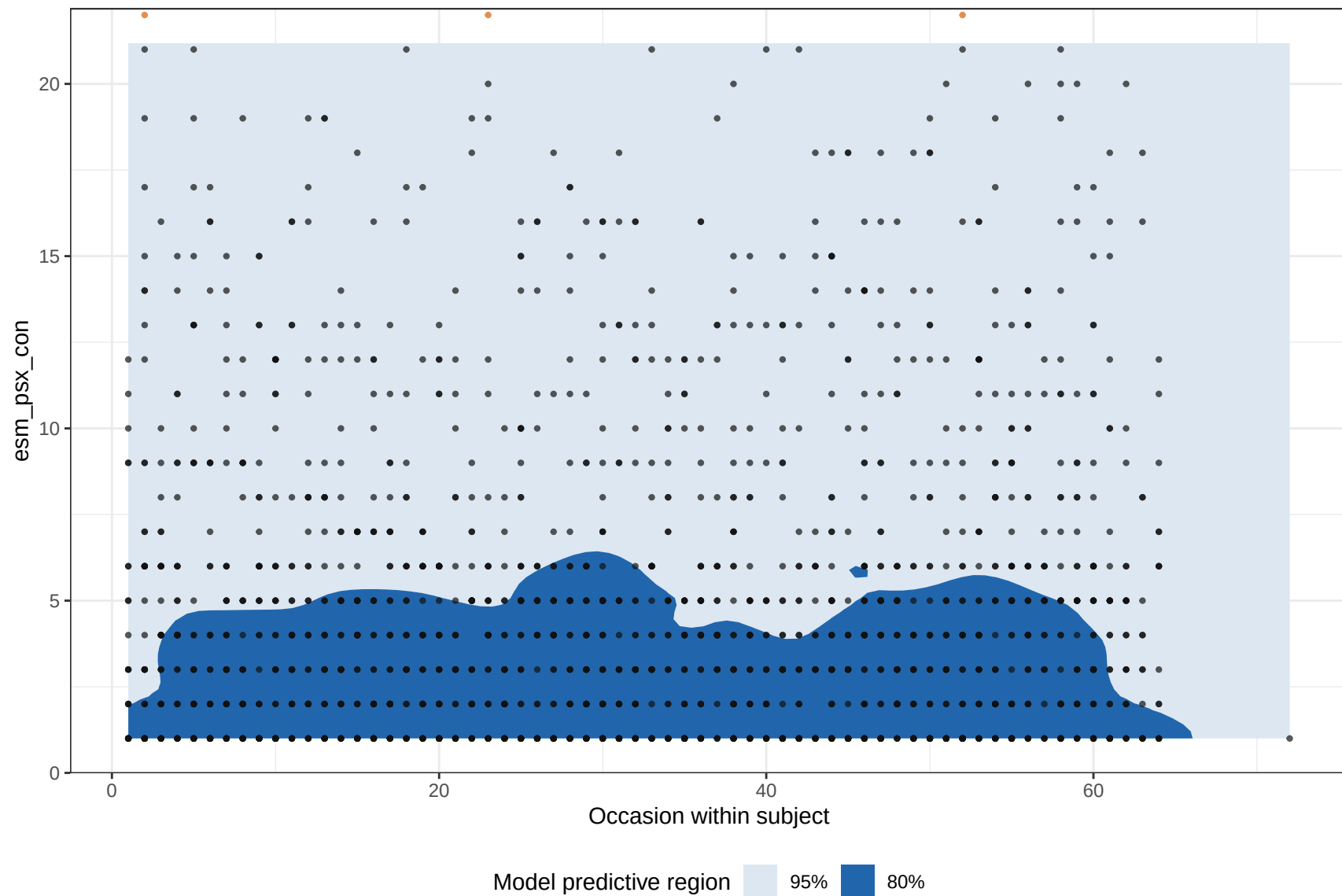
Points: observed. 0.9% fall outside the 95% region (orange); about 5% should.



Shaded: where the model puts 50, 80 and 95% of its predictive mass, from 200 draws. Points: the observed data, orange outside the 95% region. A systematic drift of the points off the regions means the trend is wrong; regions much wider or narrower than the scatter mean the variance is.

esm_psx_con

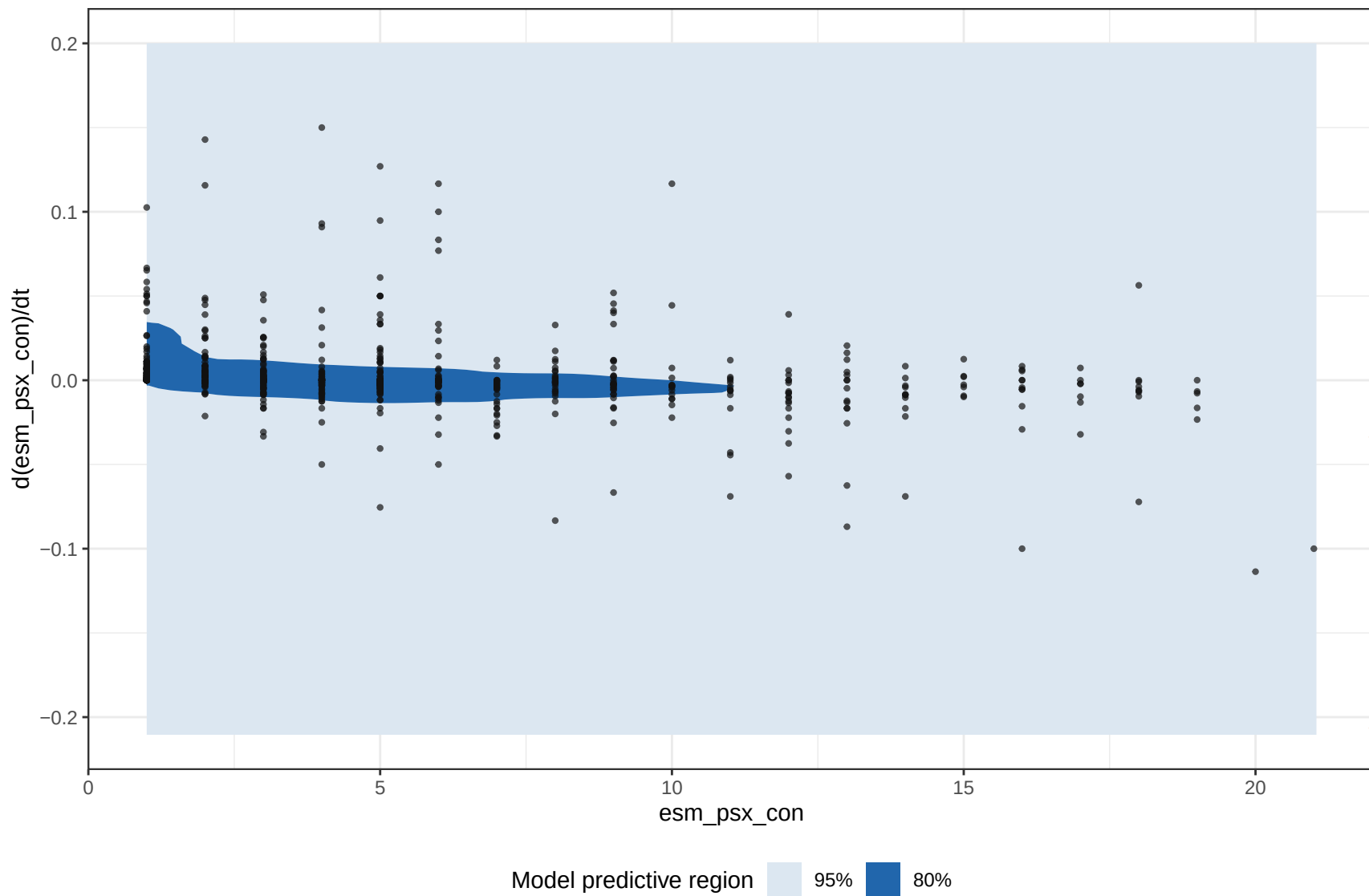
Points: observed. 0.9% fall outside the 95% region (orange); about 5% should.



As the time panel, but against occasion number within subject rather than elapsed time. Separates a misfit that follows measurement order (fatigue, practice, panel conditioning) from one that follows time.

esm_psx_con, lag 1

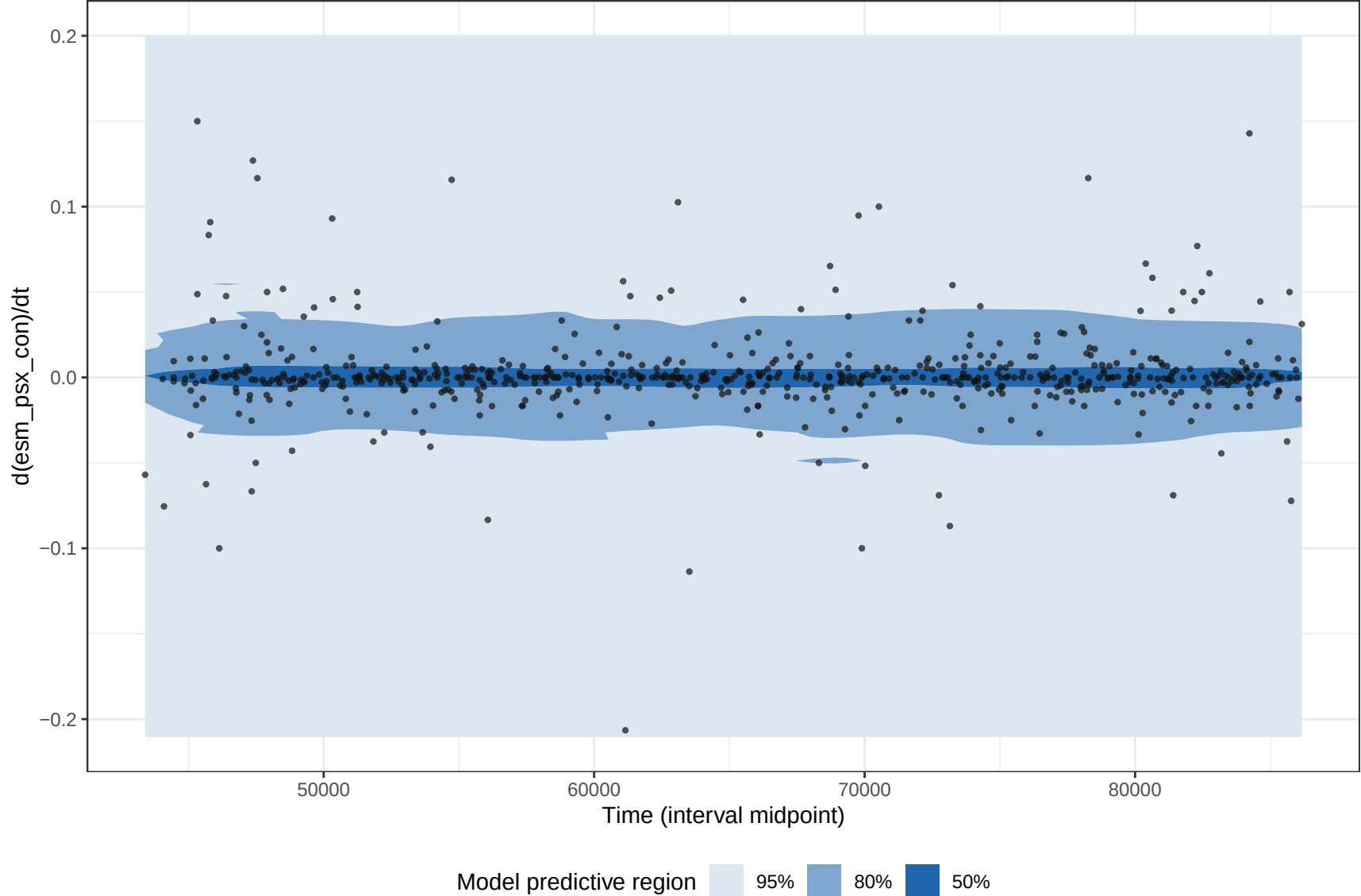
Points: observed. 1.9% fall outside the 95% region (orange); about 5% should.



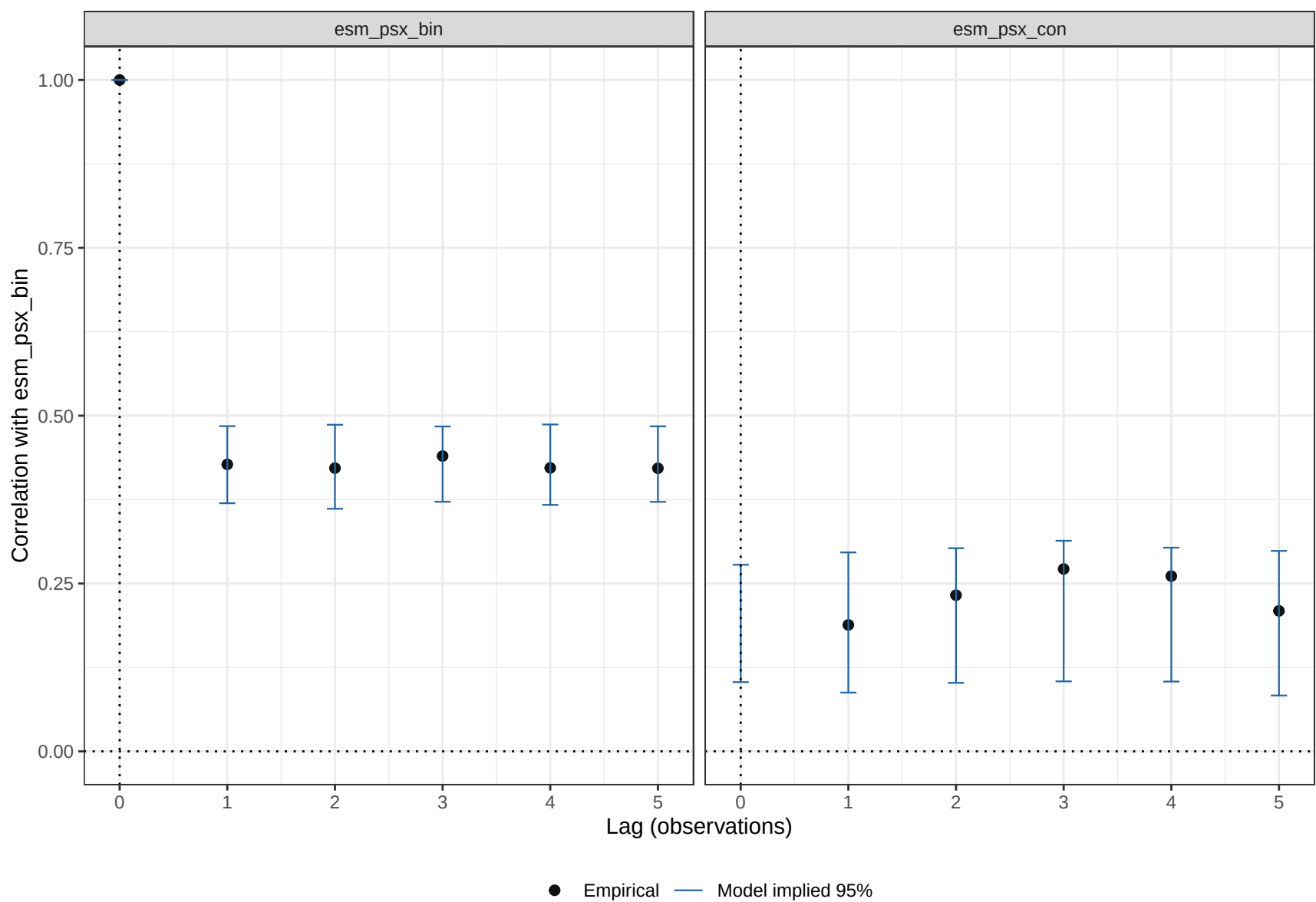
Phase plane: rate of change against the value it started from, shaded by the 50, 80 and 95% predictive regions. A linear model implies a band sloping down at the drift rate, the same everywhere. Observed points that curve, or that flatten at the extremes while the regions do not, indicate nonlinear dynamics. See `ctPhasePortrait()` for the model side of this on its own.

esm_psx_con, lag 1

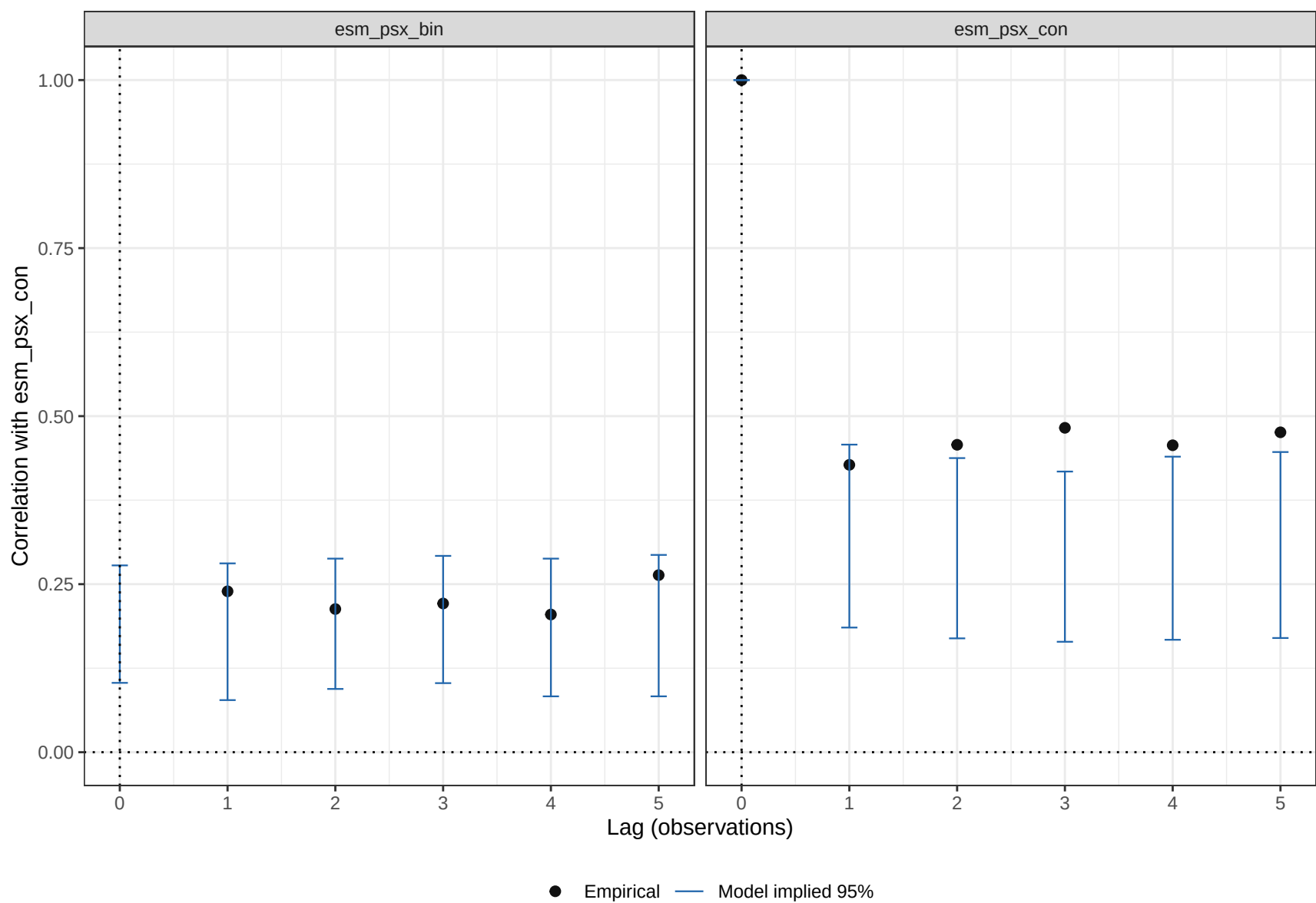
Points: observed. 0.7% fall outside the 95% region (orange); about 5% should.



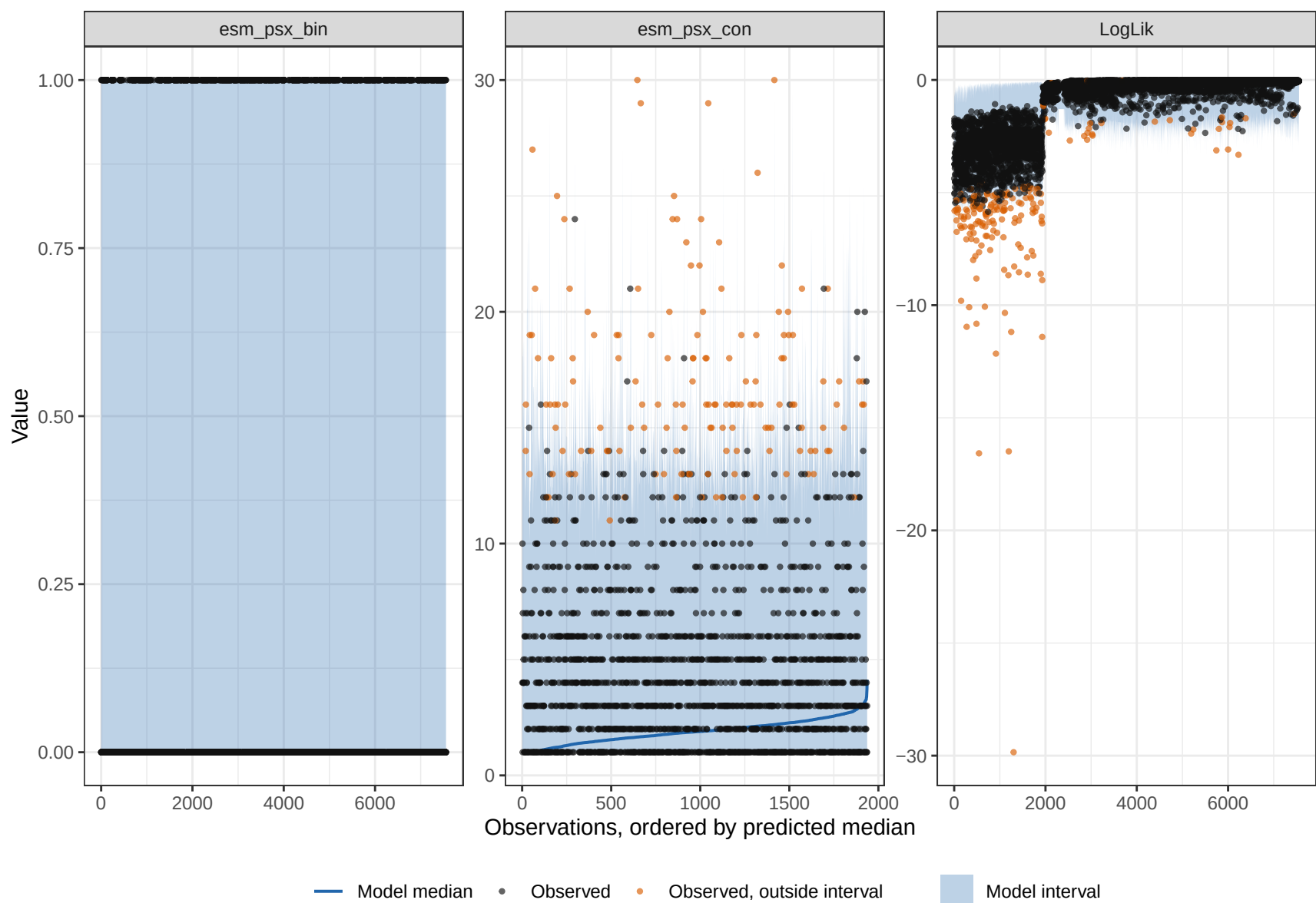
Rate of change against time, plotted at the midpoint of each interval and shaded by the 50, 80 and 95% predictive regions. A scatter that widens or narrows over time while the regions do not points at the diffusion rather than the drift.



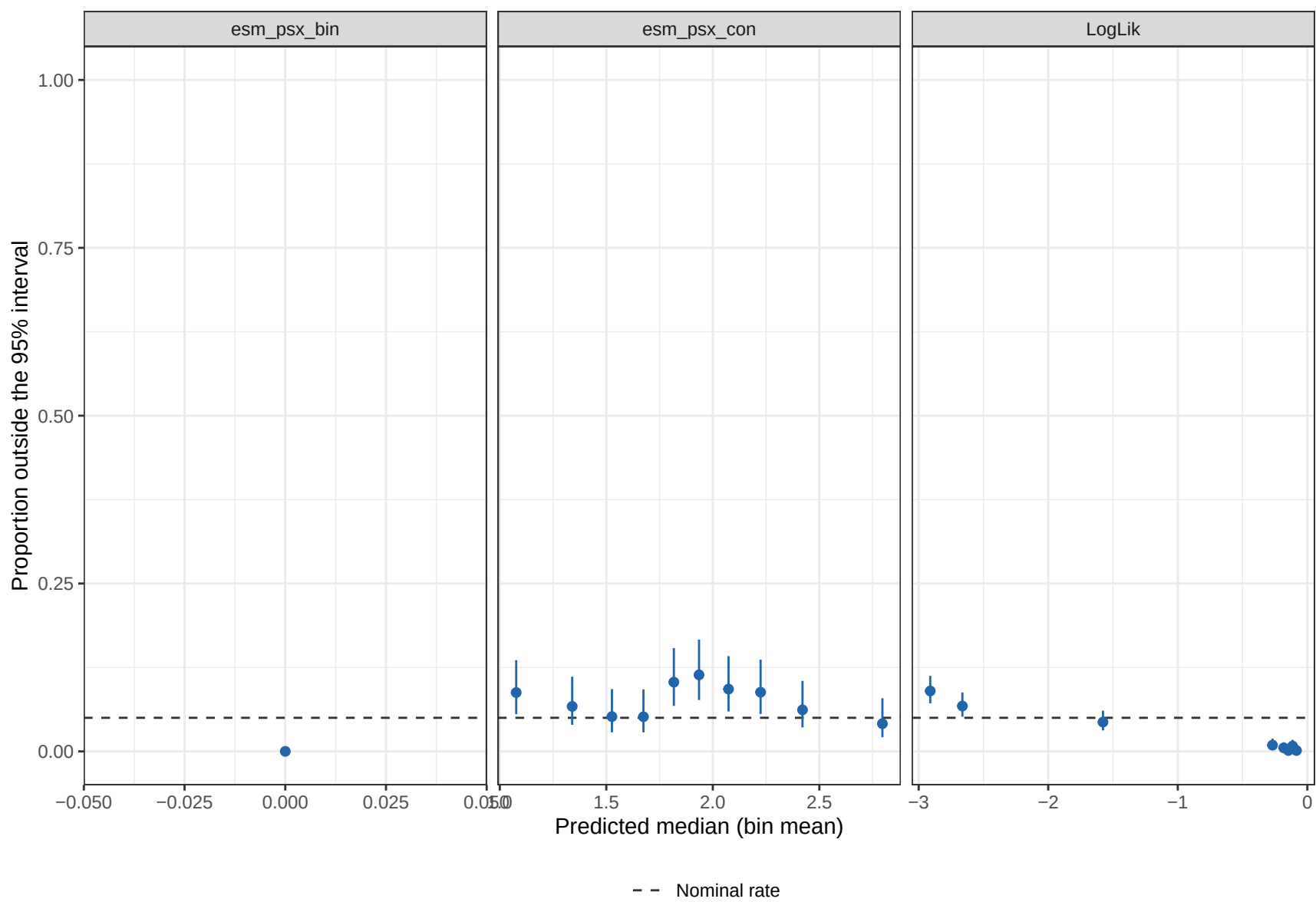
Lagged covariance or correlation of each variable with each other variable, computed across subjects and averaged over matching observation-index pairs. Points: the empirical value. Bars: the central 95% of the same quantity over 200 generated datasets. This is the panel that asks about dependence ACROSS time rather than the marginal at each time, so it is where a wrong drift or diffusion shows up even when every marginal matches. Computed by `ctFitCheckCov()`, which can be called directly for splits by group.



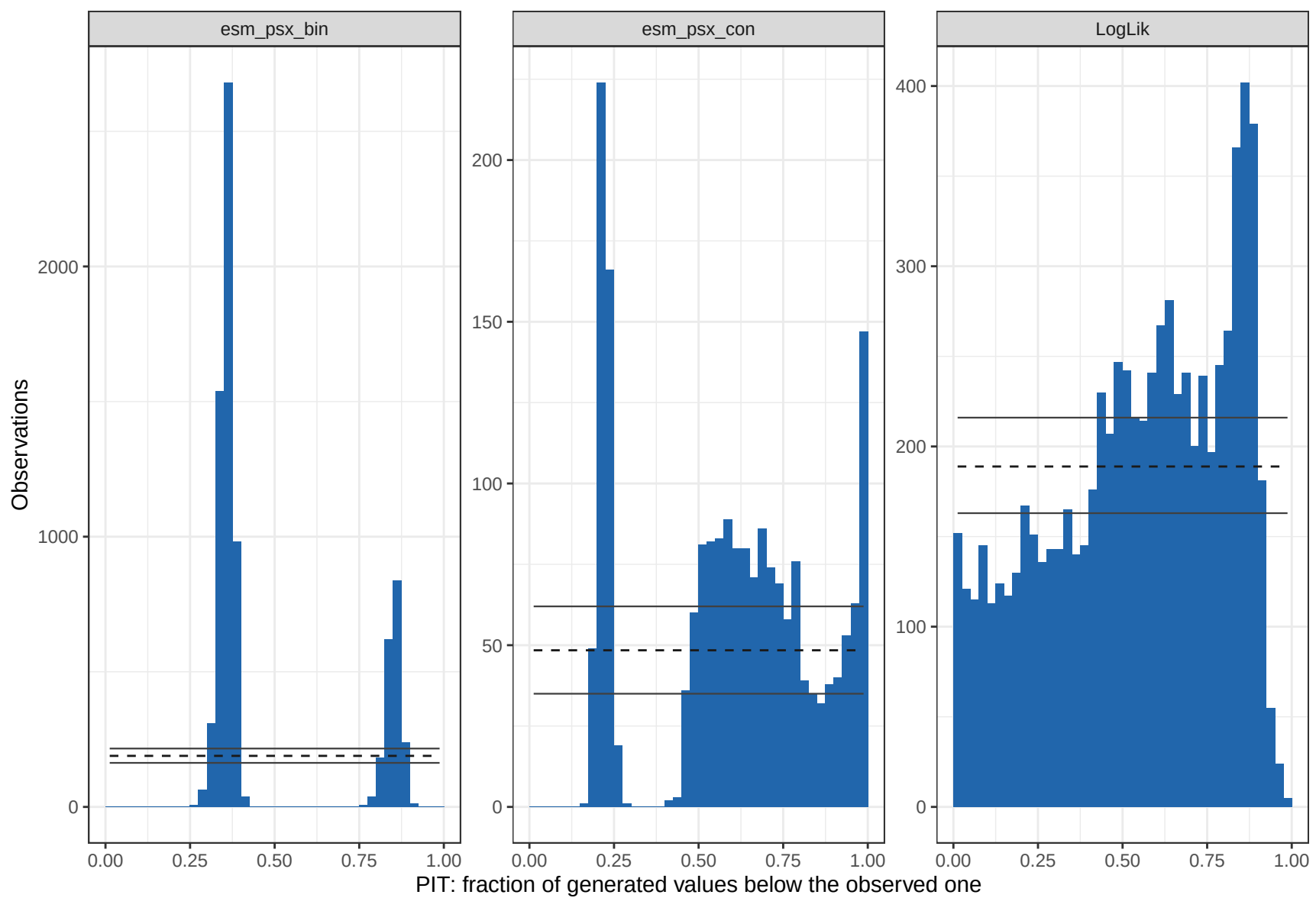
Lagged covariance or correlation of each variable with each other variable, computed across subjects and averaged over matching observation-index pairs. Points: the empirical value. Bars: the central 95% of the same quantity over 200 generated datasets. This is the panel that asks about dependence ACROSS time rather than the marginal at each time, so it is where a wrong drift or diffusion shows up even when every marginal matches. Computed by `ctFitCheckCov()`, which can be called directly for splits by group.



Each observation against the central 95% predictive interval for it, ordered along the x axis by predicted median so the interval reads as a band. Orange points fall outside their own interval; about 5% should. LogLik is each row's log-likelihood, and its distribution should match like any other quantity -- but the observed side is evaluated at the point estimate, so it carries no parameter uncertainty while the generated side does.

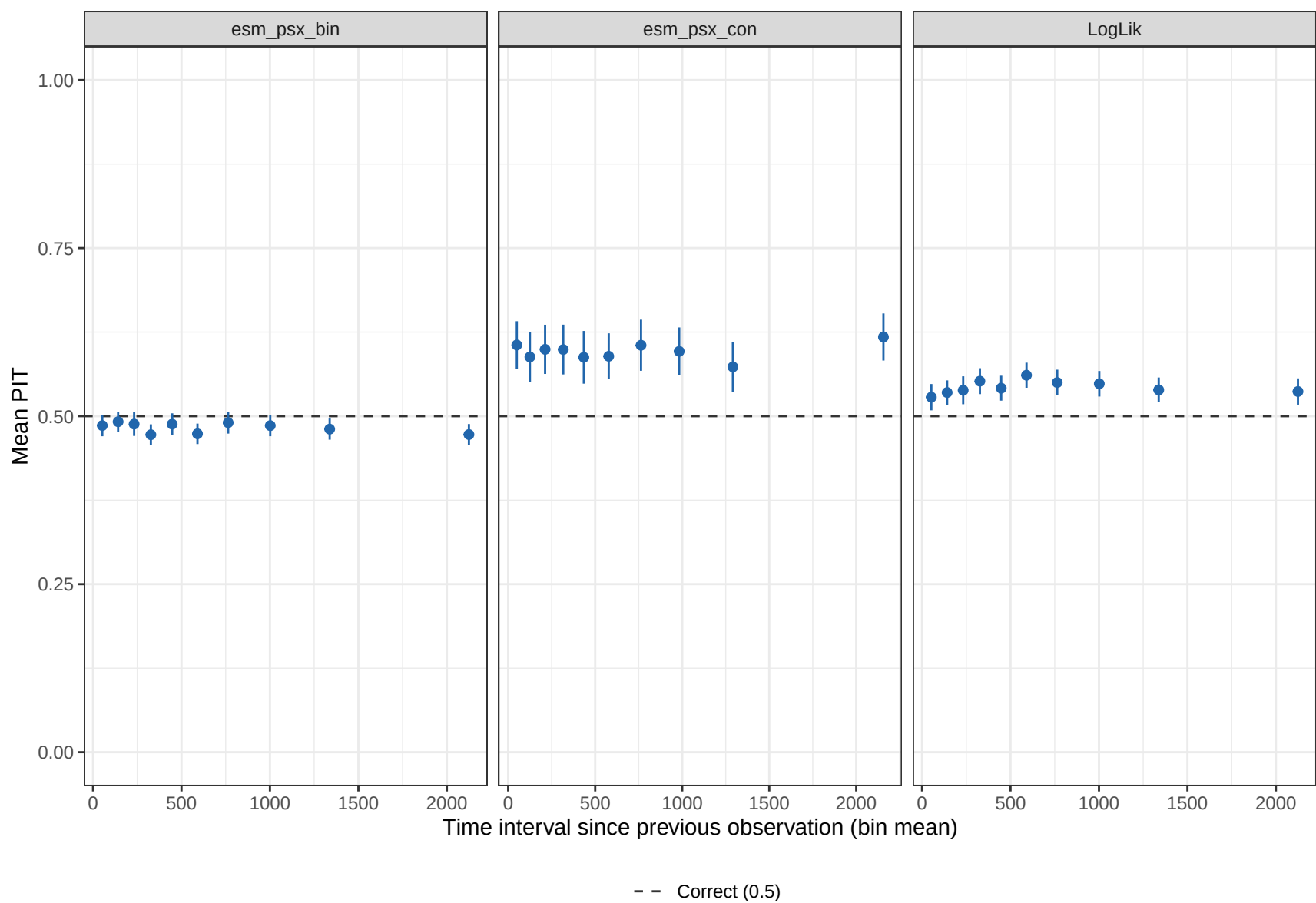


The same exceedances as a rate, in ten equal-count bins of predicted median, so a misfit concentrated among high or low predictions is visible. Dashed line: the nominal 5%. Bars: 95% Wilson interval for each bin. LogLik is each row's log-likelihood, and its distribution should match like any other quantity -- but the observed side is evaluated at the point estimate, so it carries no parameter uncertainty while the generated side does.

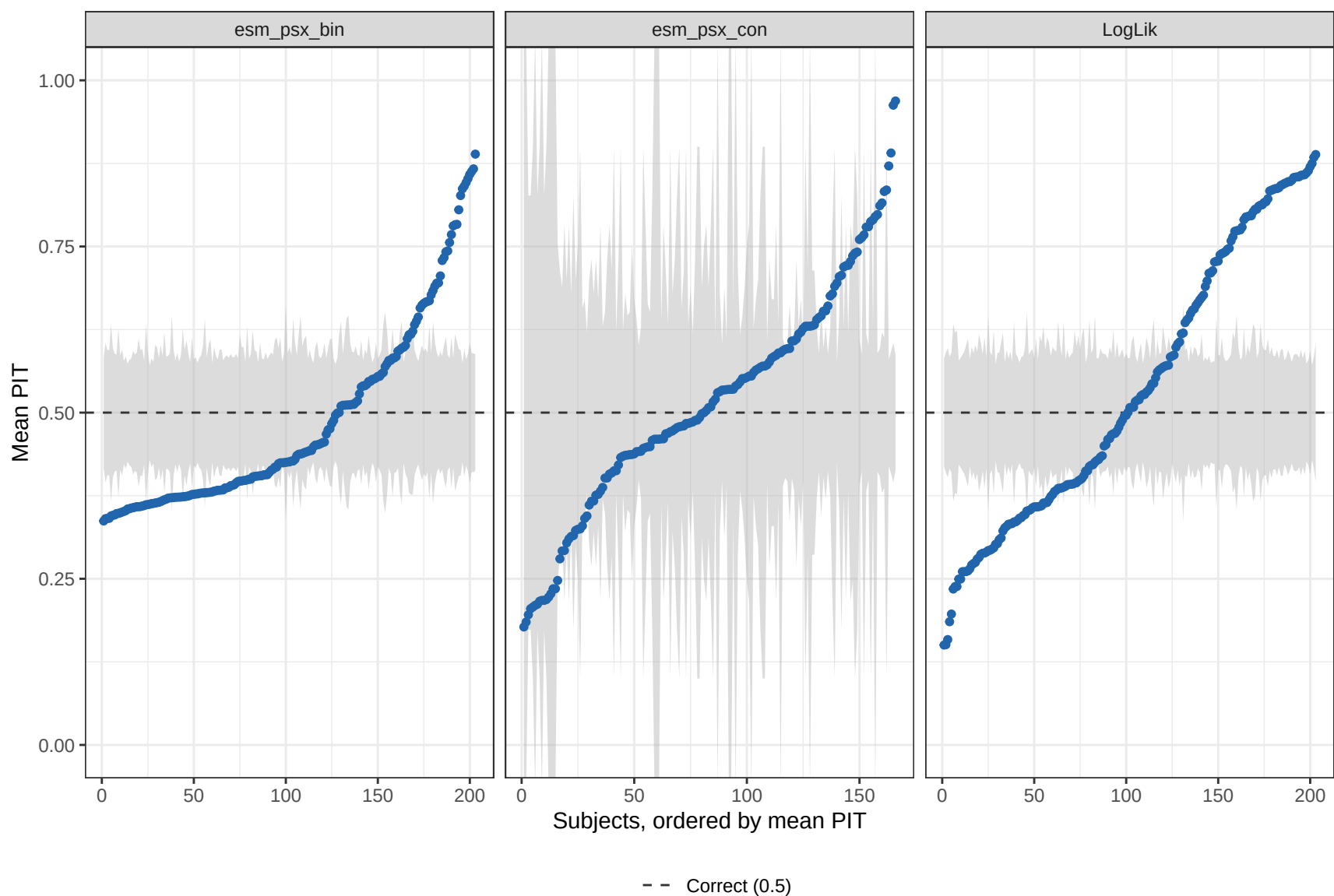


Probability integral transform -- for each observation, the fraction of the 200 generated values falling below it. Under a correctly calibrated model this is uniform. A hump in the middle means the predictive distribution is too wide, peaks at both ends mean it is too narrow, and a tilt means a location bias.

Grey lines: where 95% of bin counts should fall under uniformity. LogLik is each row's log-likelihood, and its distribution should match like any other quantity -- but the observed side is evaluated at the point estimate, so it carries no parameter uncertainty while the generated side does.

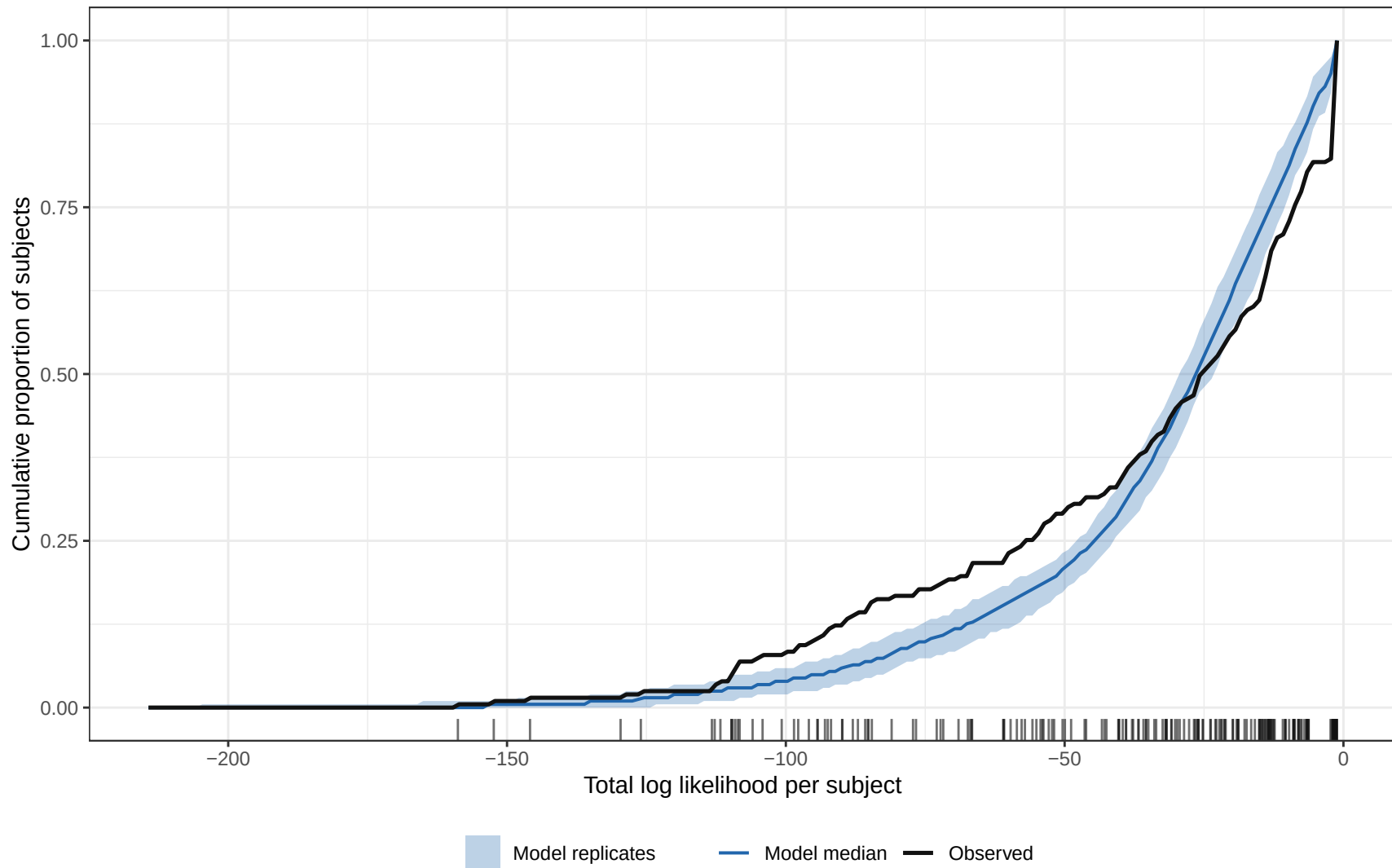


Mean PIT against the time interval since the previous observation, in equal-count bins; 0.5 is correct. A departure that depends on the interval implicates the dynamics -- drift or diffusion -- rather than the measurement model, which cannot know the interval. LogLik is each row's log-likelihood, and its distribution should match like any other quantity -- but the observed side is evaluated at the point estimate, so it carries no parameter uncertainty while the generated side does.



Mean PIT per subject, sorted; 0.5 is correct, and subjects at the ends are the ones the model fits worst. Grey band: the range expected if a subject's own observations were independent. They are not -- a random effect or a persistent state makes them agree -- so the true band is wider, and this one is a guide rather than a test. LogLik is each row's log-likelihood, and its distribution should match like any other quantity -- but the observed side is evaluated at the point estimate, so it carries no parameter uncertainty while the generated side does.

203 subjects, 200 replicate datasets. Observed curve outside the band over 42.5% of the range.



Distribution across subjects of each subject total log likelihood, as a cumulative curve. Band: the same curve from 200 replicate datasets the model generated, each with the same subjects and the same observation schedule, so it is the right reference however few subjects there are. The observed curve should sit inside it. Left of the band means subjects are less likely under the model than it expects -- systematic misfit spread across occasions; right means the model is fitting them better than it should, which usually means overfitting or a variance component absorbing the noise. The rug marks the individual subjects; CalibrationBySubject names which are which. The observed totals are evaluated at the point estimate, so they carry no parameter uncertainty while the replicates do.